

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    'State': ['Andhra Pradesh', 'Telangana', 'Tamil Nadu',
             'Karnataka', 'Kerala', 'Maharashtra',
             'Delhi', 'Punjab'],

    'Unemployment Rate (%)': [7.5, 8.2, 6.1, 5.8, 9.0, 7.2, 10.5, 6.8],

    'Employed': [500000, 450000, 620000, 580000,
                 400000, 700000, 300000, 350000],

    'Labour Participation Rate (%)': [42.5, 44.2, 41.8,
                                     40.5, 39.7, 45.1,
                                     38.9, 41.2]
}

df = pd.DataFrame(data)

print("UNEMPLOYMENT DATASET\n")
print(df)
print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nMissing Values:")
```

```
sns.heatmap(  
    df.corr(numeric_only=True),  
    annot=True,  
    cmap='coolwarm'  
)  
  
plt.title('Correlation Heatmap')  
  
plt.show()  
highest = df.loc[  
    df['Unemployment Rate (%)'].idxmax()  
]  
  
print("\nState with Highest Unemployment:\n")  
print(highest)  
  
print("\nConclusion:")  
print("Delhi has the highest unemployment rate in this dataset.")  
print("The project successfully analyzes unemployment data using Python.")
```

UNEMPLOYMENT DATASET

	State	Unemployment Rate (%)	Employed \
0	Andhra Pradesh	7.5	500000
1	Telangana	8.2	450000
2	Tamil Nadu	6.1	620000
3	Karnataka	5.8	580000
4	Kerala	9.0	400000
5	Maharashtra	7.2	700000
6	Delhi	10.5	300000
7	Punjab	6.8	350000

	Labour Participation Rate (%)
0	42.5
1	44.2
2	41.8
3	40.5
4	39.7
5	45.1
6	38.9
7	41.2

Dataset Shape:
(8, 4)

Column Names:

```
Index(['State', 'Unemployment Rate (%)', 'Employed',  
      'Labour Participation Rate (%)'],  
      dtype='object')
```

```
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())
plt.figure(figsize=(10,5))

sns.barplot(
    x='State',
    y='Unemployment Rate (%)',
    data=df
)

plt.title('State-wise Unemployment Rate')
plt.xticks(rotation=45)

plt.show()

plt.figure(figsize=(8,5))

sns.histplot(
    df['Unemployment Rate (%)'],
    kde=True
)

plt.title('Distribution of Unemployment Rate')

plt.show()

plt.figure(figsize=(8,5))

sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
```

Missing Values:

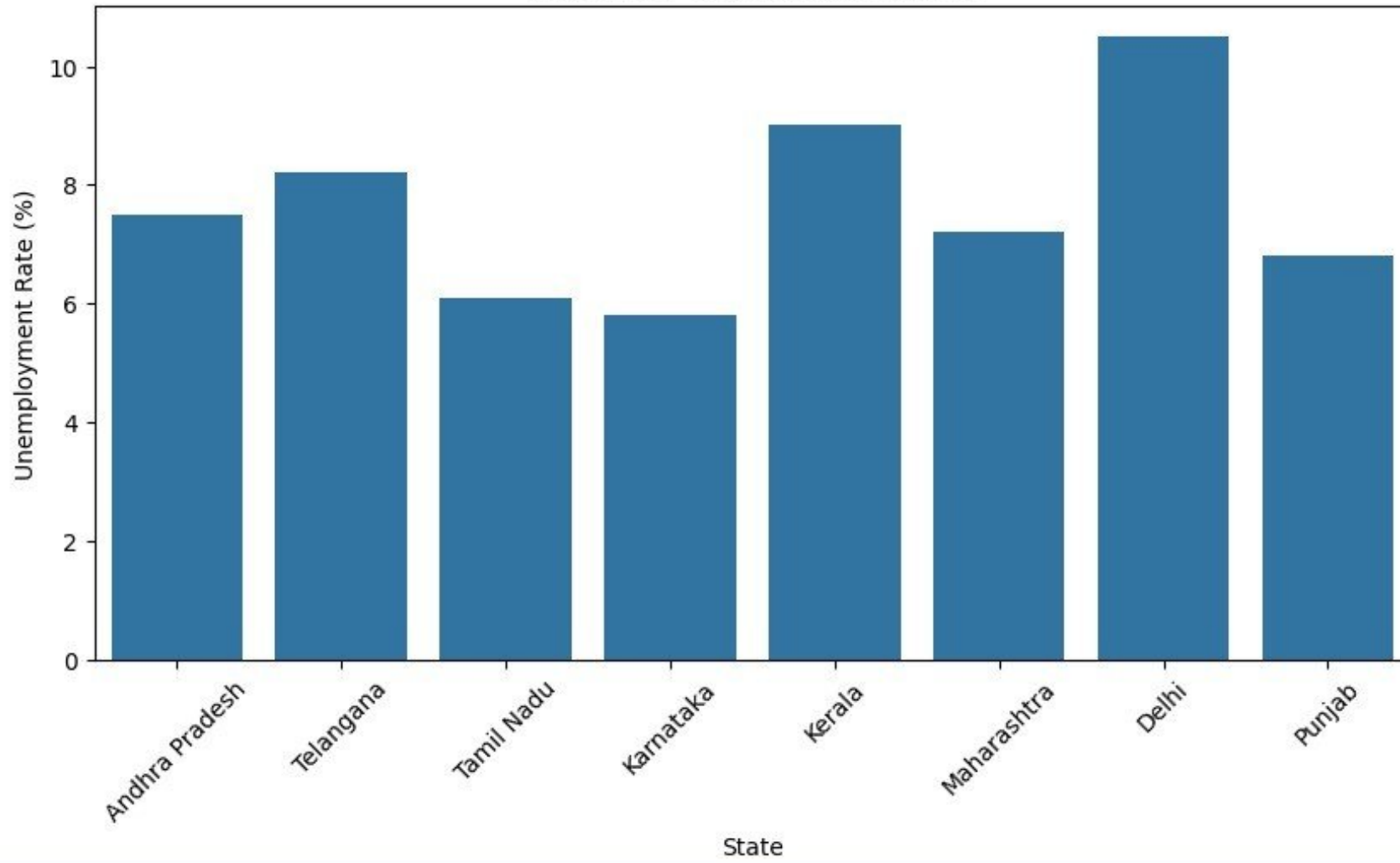
State	0
Unemployment Rate (%)	0
Employed	0
Labour Participation Rate (%)	0

dtype: int64

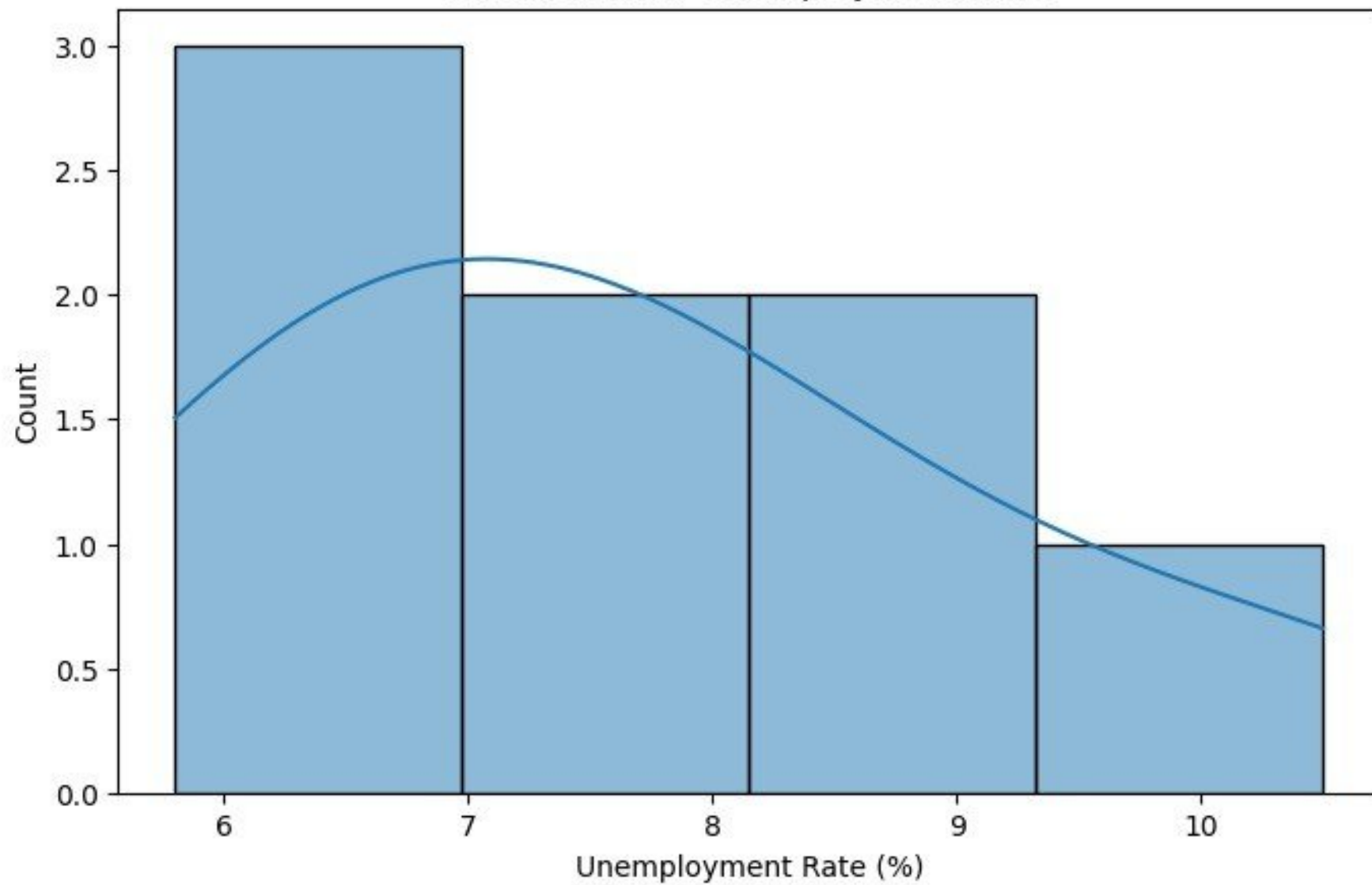
Statistical Summary:

	Unemployment Rate (%)	Employed	Labour Participation Rate (%)
count	8.000000	8.000000	8.000000
mean	7.637500	487500.00000	41.73750
std	1.559247	138641.36674	2.13738
min	5.800000	300000.00000	38.90000
25%	6.625000	387500.00000	40.30000
50%	7.350000	475000.00000	41.50000
75%	8.400000	590000.00000	42.92500
max	10.500000	700000.00000	45.10000

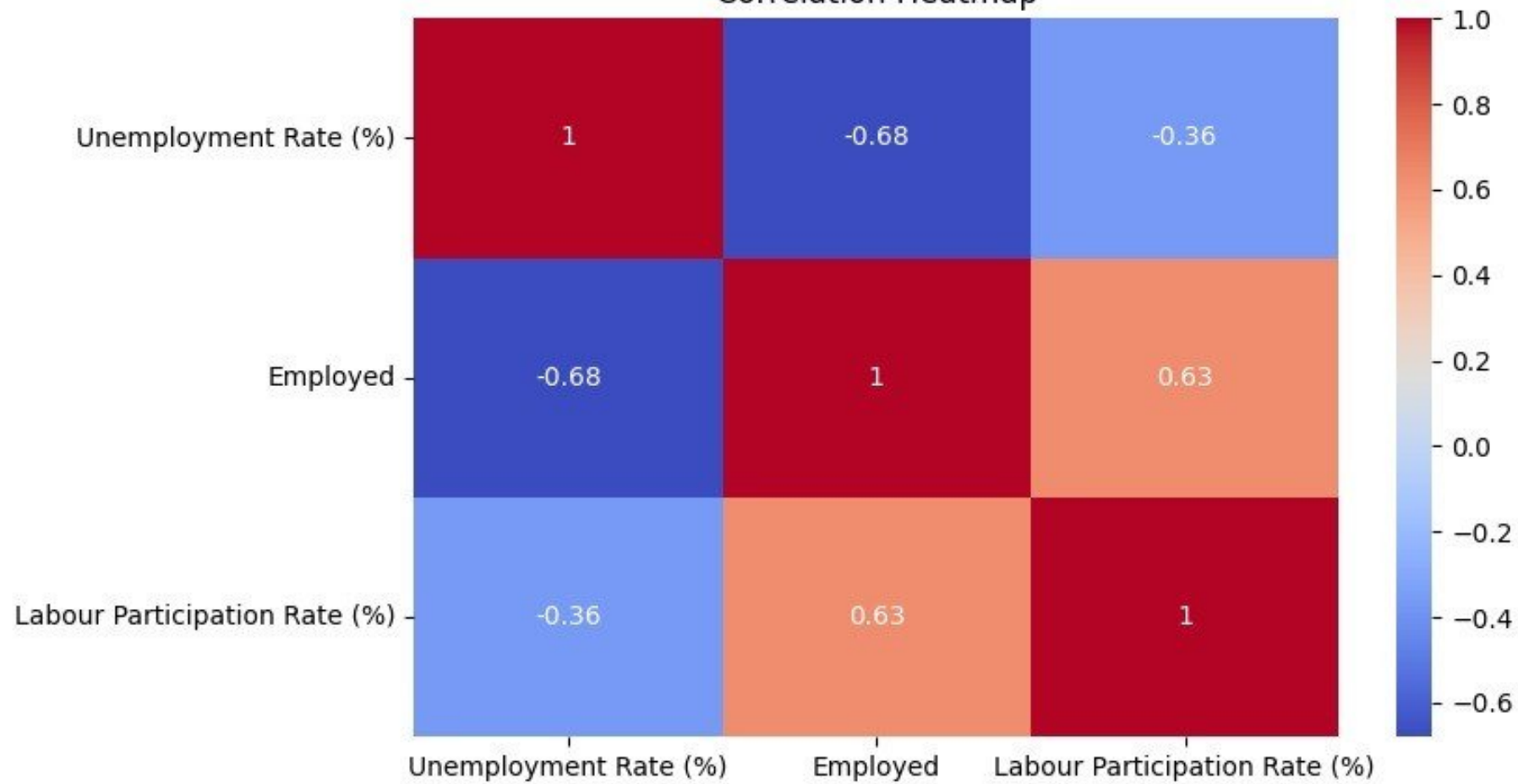
State-wise Unemployment Rate



Distribution of Unemployment Rate



Correlation Heatmap



State with Highest Unemployment:

State	Delhi
Unemployment Rate (%)	10.5
Employed	300000
Labour Participation Rate (%)	38.9

Name: 6, dtype: object

Conclusion:

Delhi has the highest unemployment rate in this dataset.

The project successfully analyzes unemployment data using Python.